Operating Systems (Sessional)

CSE-335

OS Project: CUET CSE Student Management System

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Introduction

The CUET CSE Student Management System (CLI-SMS) is a shell-based operating system project designed to digitally manage academic records for the Department of Computer Science and Engineering at Chittagong University of Engineering and Technology. The system supports the complete academic lifecycle — from student registration and batch management to course enrollment, grade entry, and cumulative GPA calculation across all eight semesters of the CSE program.

The system provides three distinct roles — Admin, Teacher, and Student — each with a dedicated dashboard and role-appropriate features. All data is persisted using plain text files, and authentication is secured using SHA-256 password hashing. This project demonstrates practical OS-level scripting skills and simulates file-based record management without the need for an external database engine.

System Requirements

The CUET CSE Student Management System is a shell-based application that runs entirely in a terminal environment. The following requirements must be met to run the system:

- Operating System: Linux, macOS, or Windows with WSL (Windows Subsystem for Linux)
- Shell: Bash version 4.0 or higher
- Required utilities: bash, awk, grep, sed, cut, sha256sum, date, mktemp
- No external database engine required — all data is stored in plain text files
- Read and write permissions in the project working directory
- Minimum screen width of 80 columns for proper table rendering

To run the system:

- Fresh start: bash main.sh
- With demo data pre-loaded: bash main.sh --demo

Learning Objectives

1. Understand the design and implementation of a multi-role, shell-based information management system.

2. Gain hands-on experience with Bash scripting for real-world application logic including menus, input validation, and file I/O.

3. Learn to implement secure authentication using SHA-256 password hashing and session management through shell environment variables.

4. Practice structured data storage and retrieval using pipe-delimited flat files as a lightweight alternative to relational databases.

5. Demonstrate role-based access control (RBAC) by implementing separate panels with distinct privileges for Admin, Teacher, and Student roles.

6. Develop skills in academic record tracking including multi-semester enrollment, grade management, and weighted CGPA calculation.

7. Experience the full software development lifecycle — from requirements analysis and modular design to implementation and testing.

Manual

I. User Roles

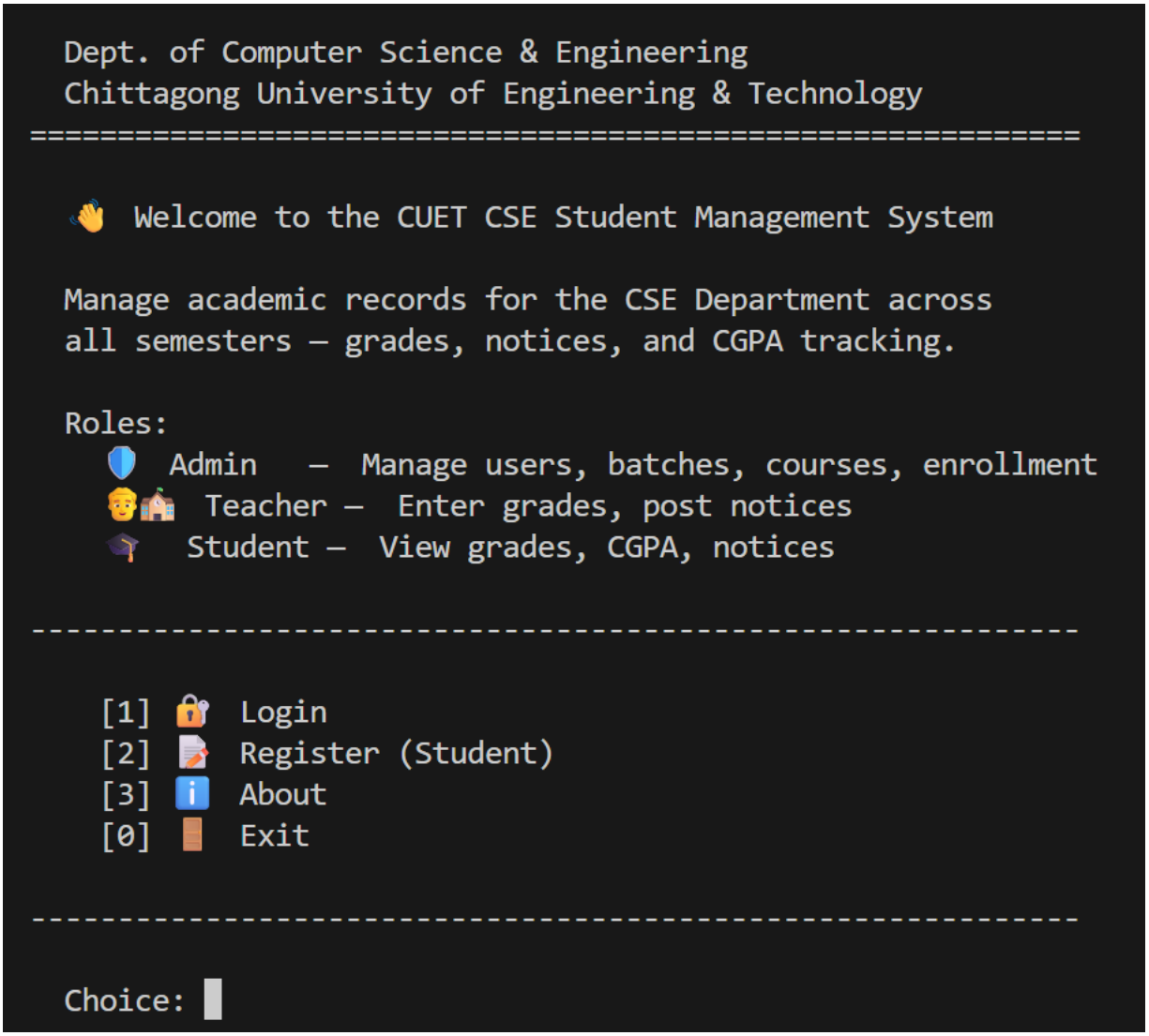

Figure 1: Main Menu — Landing Page of the CUET CSE SMS

The system defines three user roles, each with a dedicated panel and a specific set of features:

1. Admin : The Admin panel provides full control over the system. The key features available to the administrator are:

- User Management (Add / Delete / Reset Password for Students, Teachers, Admins)
- Pending Registration Management (Approve or Reject student self-registrations)
- Batch Management (Create, Advance, Graduate batches)
- Course and Teacher Assignment

- Enrollment Management (Auto-enroll batches per semester)
- Reports (Grade Sheets, Full Batch Academic Reports)
- Activity Log Viewer

2. Teacher : The Teacher panel allows educators to manage grades and communicate with students within their assigned courses. The key features available to the teacher are:

- View Assigned Courses
- View Enrolled Students per Course
- Enter and Update Student Grades
- View Grade Sheet with Class Average
- Post Course Notices
- View Course Notices
- Change Password

3. Student: The Student panel allows students to track their academic progress and stay informed about course updates. The key features available to the student are:

- Self-Registration with Payment Transaction Verification
- Dashboard (Batch info, Current CGPA, Completed Credits)
- View Current Semester Courses
- View Current Semester Grades and Semester GPA
- View Full Academic Record and Cumulative GPA (CGPA)
- View Course Notices
- Change Password

II. Admin Manual

Admin Login

- Admin enters their username and password at the Login screen.
- The password is SHA-256 hashed and compared against the stored hash in db/users.txt.
- On success, a welcome message is displayed and the Admin Panel is loaded.

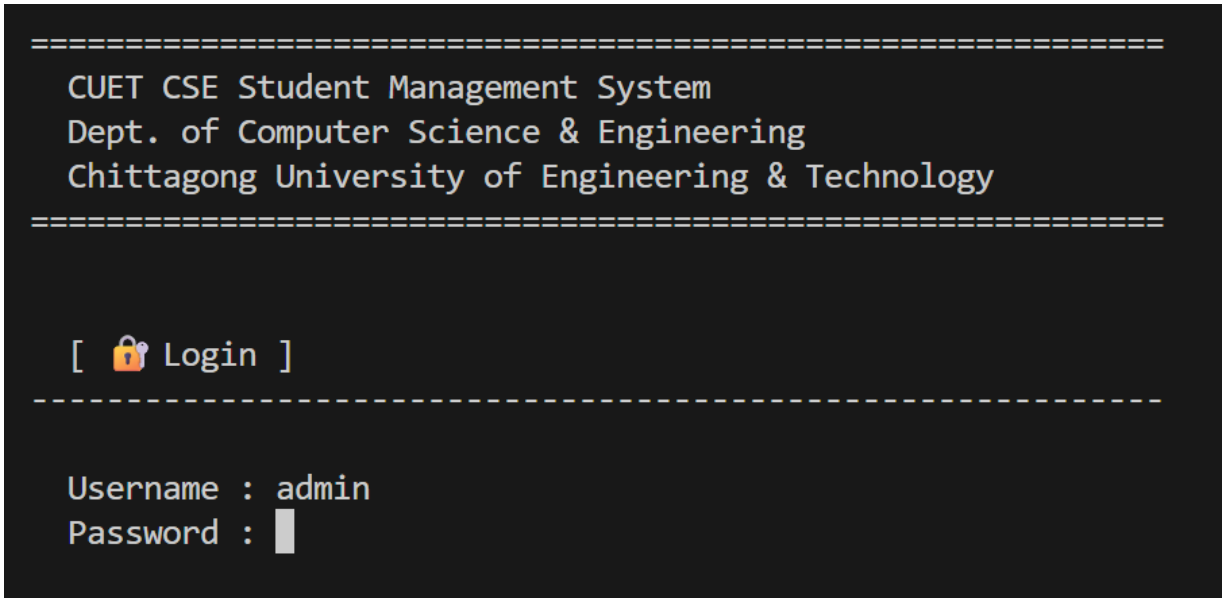

Figure 2: Admin Login Screen

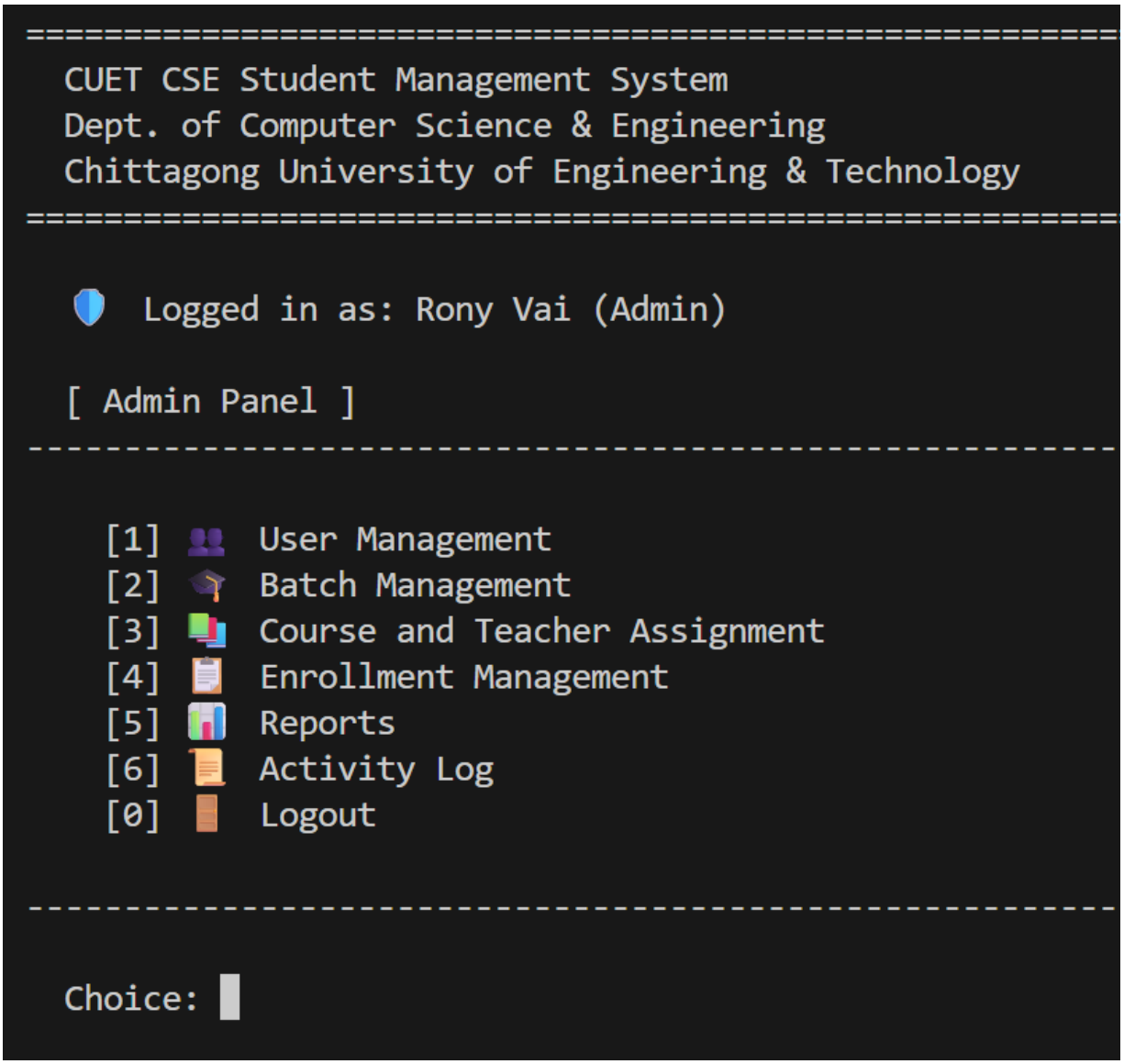

Figure 3: Admin Panel Main Menu

1. User Management

1.1. Add Student

- **Function:** Create a new student account and assign them to a batch.
- **Process:**
  - Admin enters a unique username (letters, numbers, underscores only).
  - Admin enters the student's full name.
  - Admin selects the target batch from the batch picker.
  - Admin sets a temporary password (minimum 6 characters).

○ The account is created with a force-password-change flag; the student must change their password on first login.

```
============================================================
  CUET CSE Student Management System
  Dept. of Computer Science & Engineering
  Chittagong University of Engineering & Technology
============================================================


  [ Add student ]
------------------------------------------------------------

  Username   : Tahsin028
  Full Name  : Tahsin Chowdhury

  Select Batch
------------------------------------------------------------
    [1] CSE2024 (Level-2 Term-1, active)
    [0] Cancel
------------------------------------------------------------

  Choice: 1
  Temp Password :

  ✅ User 'Tahsin028' created.
```

Figure 4: Add Student Interface

1.2. Add Teacher

- **Function:** Create a new teacher account.
- **Process:**
  - Admin enters a unique username and the teacher's full name.
  - Admin sets a temporary password.
  - Account is created with a force-password-change flag.

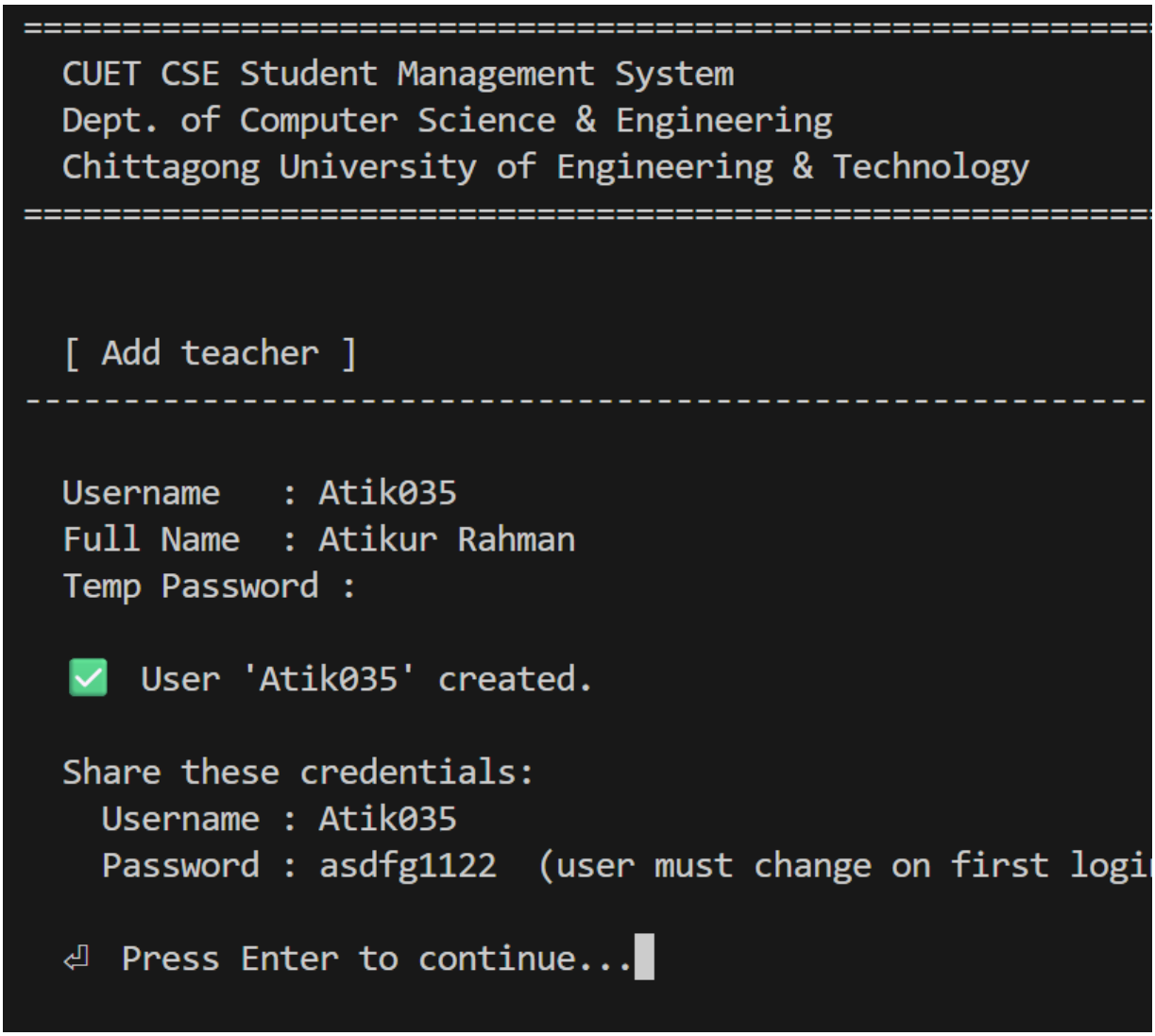

Figure 5: Add Teacher Interface

1.3. Add Admin

- **Function:** Create an additional administrator account.
- **Process:**
  - Same flow as adding a teacher — username, full name, and temporary password.
  - The new admin gains full admin panel access on first login after changing their password.

1.4. List All Users

- **Function:** Display a table of all registered users in the system.
- **Process:**
  - Reads from db/users.txt and prints all users.
  - Columns displayed: Username, Full Name, Role, Batch.

```
  [ All Users ]
------------------------------------------------------------

  Username        Full Name                    Role       Batch
------------------------------------------------------------
  admin           Rony Vai                     admin      -
  prof_karim      Dr. Abdul Karim              teacher    -
  prof_rahman     Dr. Md. Rashidur Rahman      teacher    -
  sara2024        Sara Islam                   student    CSE2024
  karim2024       Karim Hossain                student    CSE2024
  rhrony05        Muhammad Rony                student    CSE2024
  rony2024        Rony Ahmed                   student    CSE2024
  rafNsane        Rafsane Ron                  student    CSE2024
  Tahsin028       Tahsin Chowdhury             student    CSE2024
  Atik035         Atikur Rahman                teacher    -

  ⏎  Press Enter to continue...
```

Figure 6: All Users Table

1.5. Delete User

- **Function:** Remove a user account from the system.
- **Process:**
  - Admin selects a user from the picker.
  - An admin cannot delete their own account.
  - Admin confirms the deletion.
  - The user record is removed from db/users.txt.

```
  Select User
------------------------------------------------------------
    [1] admin - Rony Vai (admin)
    [2] prof_karim - Dr. Abdul Karim (teacher)
    [3] prof_rahman - Dr. Md. Rashidur Rahman (teacher)
    [4] sara2024 - Sara Islam (student, CSE2024)
    [5] karim2024 - Karim Hossain (student, CSE2024)
    [6] rhrony05 - Muhammad Rony (student, CSE2024)
    [7] rony2024 - Rony Ahmed (student, CSE2024)
    [8] rafNsane - Rafsane Ron (student, CSE2024)
    [9] Tahsin028 - Tahsin Chowdhury (student, CSE2024)
    [10] Atik035 - Atikur Rahman (teacher)
    [0] Cancel
------------------------------------------------------------

  Choice: 10
  User : Atik035 | Atikur Rahman | teacher

  Delete this user? (y/n): y
  ✅ User 'Atik035' deleted.
```

Figure 7: Delete User Confirmation

1.6. Reset User Password

- **Function:** Set a new password for any user and force them to change it on next login.
- **Process:**
  - Admin selects a user from the picker.
  - Admin sets a new password (minimum 6 characters).
  - The password is SHA-256 hashed and stored; the force-password-change flag is set to 1.

```
  Select User
------------------------------------------------------------
    [1] admin - Rony Vai (admin)
    [2] prof_karim - Dr. Abdul Karim (teacher)
    [3] prof_rahman - Dr. Md. Rashidur Rahman (teacher)
    [4] sara2024 - Sara Islam (student, CSE2024)
    [5] karim2024 - Karim Hossain (student, CSE2024)
    [6] rhrony05 - Muhammad Rony (student, CSE2024)
    [7] rony2024 - Rony Ahmed (student, CSE2024)
    [8] rafNsane - Rafsane Ron (student, CSE2024)
    [9] Tahsin028 - Tahsin Chowdhury (student, CSE2024)
    [0] Cancel
------------------------------------------------------------

  Choice: 9
  New Password:
  ✅ Password reset for 'Tahsin028'. They must change it on next login.
```

Figure 8: Reset User Password Interface

1.7. Pending Registrations

- **Function:** Review and approve or reject student self-registrations.
- **Process:**
  - Displays a list of all registrations with status 'pending'.
  - Admin selects a registration to review — shows username, full name, batch, payment transaction ID, and submission time.
  - Admin can Approve (creates the student account) or Reject (marks the registration as rejected).
  - Approved students can immediately log in with the password they set during registration.

```
  Pending Registrations to Process
------------------------------------------------------------
    [1] newstudent12 - Test Student (CSE2024) — TRX: TRXX123456
    [0] Cancel
------------------------------------------------------------

  Choice:
```

Figure 9: Pending Registration Review Interface

2. Batch Management

2.1. Create New Batch

- **Function:** Register a new student batch in the system.
- **Process:**
  - Admin provides a unique Batch ID (e.g., CSE2025) and admission year.
  - The batch is initialized at Level-1 Term-1 with status 'active'.
  - Admin must then use Enrollment Management to enroll students into courses.

```
  [ Create New Batch ]
------------------------------------------------------------

  Batch ID (e.g. CSE2025): CSE2022
  Admission Year   : 2022
  ✅  Batch 'CSE2022' created (Level-1 Term-1, active).
  Next: use Enrollment Management to enroll this batch.
```

Figure 10: Create New Batch Interface

2.2. Advance Batch to Next Semester

- **Function:** Progress a batch from its current semester to the next.
- **Process:**
  - Admin selects a batch from the picker.
  - System shows the current semester and the target semester for confirmation.
  - On confirmation, all active enrollments for that batch are locked as 'completed'.
  - The batch semester counter is incremented (L1T1 → L1T2 → L2T1 → ... → L4T2).
  - At Level-4 Term-2, the admin is offered the option to graduate the batch (status set to 'completed').

```
  Select Batch
------------------------------------------------------------
    [1] CSE2024 (Level-2 Term-1, active)
    [2] CSE2022 (Level-1 Term-1, active)
    [0] Cancel
------------------------------------------------------------

  Choice: 1
  Current : Level-2 Term-1
  New     : Level-2 Term-2

  Advance batch 'CSE2024'? (y/n): y
  ✅ Batch 'CSE2024' advanced to Level-2 Term-2.
  Previous semester enrollments are now locked.
  Use Enrollment Management to enroll for the new semester.
```

Figure 11: Advance Batch Confirmation Screen

2.3. List All Batches

- **Function:** Display all batches with their current semester and status.
- **Process:**
  - Reads from db/batches.txt and prints a summary table.
  - Columns: Batch ID, Admission Year, Current Semester, Status.

```
  [ All Batches ]
------------------------------------------------------------

  Batch ID     Year   Current Semester     Status
------------------------------------------------------------
  CSE2024      2024   Level-2 Term-2       active
  CSE2022      2022   Level-1 Term-1       active

  ⏎ Press Enter to continue...
```

Figure 12: All Batches Table

2.4. View Students in Batch

- **Function:** List all students belonging to a selected batch.
- **Process:**
  - Admin selects a batch from the picker.
  - Displays all student usernames and full names in that batch.
  - Shows total student count.

```
  Select Batch
-------------------------------------------------------
    [1] CSE2024 (Level-2 Term-2, active)
    [2] CSE2022 (Level-1 Term-1, active)
    [0] Cancel
-------------------------------------------------------

  Choice: 1

  Batch: CSE2024  |  Level-2 Term-2

  Username        Full Name
-------------------------------------------------------
  sara2024        Sara Islam
  karim2024       Karim Hossain
  rhrony05        Muhammad Rony
  rony2024        Rony Ahmed
  rafNsane        Rafsane Ron
  Tahsin028       Tahsin Chowdhury

  Total: 6 student(s)
```

Figure 13: View Students in Batch

3. Course and Teacher Assignment

3.1. View Curriculum

- **Function:** Display the full pre-defined CUET CSE 8-semester curriculum.
- **Process:**
  - Reads from db/courses.txt and groups courses by semester.

- For each semester, displays: Course Code, Course Name, Credits, Type (Theory/Sessional).
- Covers all 48 courses across Level-1 Term-1 through Level-4 Term-2.

```
  ── Level-4 Term-1 ──
  Code       Course Name                              Credits  Type
-------------------------------------------------------------
  CSE441     Machine Learning                         3.00     theory
  CSE443     Information Security                     3.00     theory
  CSE445     Distributed Systems                      3.00     theory
  CSE447     Advanced Database Systems                3.00     theory
  CSE449     Elective I                               3.00     theory
  CSE442     Machine Learning Sessional               1.50     sessional
  CSE444     Information Security Sessional           1.50     sessional

  ── Level-4 Term-2 ──
  Code       Course Name                              Credits  Type
-------------------------------------------------------------
  CSE451     Thesis and Project                       6.00     theory
  CSE453     Elective II                              3.00     theory
  CSE455     Elective III                             3.00     theory
  CSE457     Professional Ethics and Society          2.00     theory
```

Figure 14: Curriculum View — All 8 Semesters

3.2. Assign Teacher to Course

- **Function:** Link a teacher to a course they are responsible for teaching.
- **Process:**
  - Admin selects a teacher from the user picker (filtered by role: teacher).
  - Admin selects a course from the course picker.
  - The assignment is saved to db/teacher_courses.txt.
  - Duplicate assignments are prevented.

```
  Select User
------------------------------------------------------------
    [1] prof_karim - Dr. Abdul Karim (teacher)
    [2] prof_rahman - Dr. Md. Rashidur Rahman (teacher)
    [0] Cancel
------------------------------------------------------------

  Choice: 2

  Select Course
------------------------------------------------------------
    [1] [CSE141] Structured Programming
    [2] [EE181] Basic Electrical Engineering
    [3] [MATH141] Differential and Integral Calculus
    [4] [PHY141] Physics I
    [5] [HUM141] English
    [6] [CSE100] Computer Fundamentals and Ethics
    [7] [CSE142] Structured Programming Sessional
    [8] [EE182] Basic Electrical Engineering Sessional
```

Figure 15: Assign Teacher to Course Interface

3.3. Remove Teacher from Course

- **Function:** Unlink a teacher from a course they are currently assigned to.
- **Process:**
  - Admin selects a teacher, then selects one of their currently assigned courses.
  - The assignment record is removed from db/teacher_courses.txt.

```
  Select User
------------------------------------------------------------
    [1] prof_karim - Dr. Abdul Karim (teacher)
    [2] prof_rahman - Dr. Md. Rashidur Rahman (teacher)
    [0] Cancel
------------------------------------------------------------

  Choice:
```

Figure 16: Remove Teacher Assignment Interface

3.4. View All Teacher Assignments

- **Function:** Display a table of all current teacher-to-course assignments.
- **Process:**
  - Reads from db/teacher_courses.txt and resolves course names.
  - Columns: Teacher Username, Course Code, Course Name.

```
  [ Teacher Course Assignments ]
------------------------------------------------------------

  Teacher            Code       Course Name
------------------------------------------------------------
  prof_karim         CSE141     Structured Programming
  prof_karim         CSE142     Structured Programming Sessional
  prof_rahman        MATH141    Differential and Integral Calculus
  prof_rahman        CSE100     Computer Fundamentals and Ethics
  prof_karim         CSE143     Object Oriented Programming
  prof_karim         CSE144     OOP Sessional
  prof_rahman        MATH143    Coordinate Geometry and Vector Analysis
  prof_karim         CSE241     Data Structures
  prof_karim         CSE242     Data Structures Sessional
  prof_karim         CSE243     Digital Electronics and Pulse Techniques
  prof_karim         CSE244     Digital Electronics Sessional
  prof_rahman        MATH241    Complex Variable, Laplace and Fourier
  prof_rahman        CSE245     Discrete Mathematics
```

Figure 17: All Teacher Assignments Table

4. Enrollment Management

4.1. Auto-Enroll Batch for Current Semester

- **Function:** Automatically enroll all students in a batch into all courses of their current semester.
- **Process:**
  - Admin selects a batch from the picker.
  - System retrieves all students in the batch and all courses for the batch's current level and term.
  - For each student-course pair not already enrolled, a new enrollment record is created with status 'active'.
  - Displays enrollment summary: number of students, courses, and new records created.

```
  Select Batch
-----------------------------------------------
    [1] CSE2024 (Level-2 Term-2, active)
    [2] CSE2022 (Level-1 Term-1, active)
    [0] Cancel
-----------------------------------------------

  Choice:
```

Figure 18: Auto-Enroll Batch Confirmation and Summary

4.2. View Enrollments for a Batch

- **Function:** Display all enrollment records for a selected batch, organized by semester.
- **Process:**
    - Admin selects a batch.
    - Enrollments are grouped and displayed by semester (Level-Term).
    - Each row shows: Student Username, Course Code, Course Name.
    - Completed semesters are clearly marked as [completed].

```
  Select Batch
------------------------------------------------------------
    [1] CSE2024 (Level-2 Term-2, active)
    [2] CSE2022 (Level-1 Term-1, active)
    [0] Cancel
------------------------------------------------------------

  Choice: 1

  Enrollments for batch: CSE2024

  --- Level-1 Term-1  [completed] ---
  Student         Code       Course
------------------------------------------------------------
  rony2024        CSE141     Structured Programming
  rony2024        EE181      Basic Electrical Engineering
  rony2024        MATH141    Differential and Integral Calculus
  rony2024        PHY141     Physics I
  rony2024        HUM141     English
```

Figure 19: Enrollment Records View

5. Reports

5.1. Grade Sheet by Course

- **Function:** Display all student grades for a selected course.
- **Process:**
  - Admin selects a course from the picker.
  - Reads from courses/{CODE}/grades.txt.
  - Displays: Username, Batch, Grade, GPA Points, Timestamp.

```
Choice: 1

Course: [CSE141] Structured Programming

Username        Batch        Grade   Points  Timestamp
-------------------------------------------------------
rony2024        CSE2024      A+      4.00    2026-04-18 12:29:12
sara2024        CSE2024      A+      4.00    2026-04-18 12:29:13
karim2024       CSE2024      B+      3.25    2026-04-18 12:29:13
```

Figure 20: Grade Sheet by Course Report

5.2. Full Batch Academic Report

- **Function:** Generate a complete academic transcript for every student in a batch.
- **Process:**
  - Admin selects a batch.
  - For each student, lists all completed course enrollments with grade and GPA points.
  - Calculates and displays the student's total completed credits and Cumulative GPA (CGPA).
  - CGPA is computed as: CGPA = Σ(Grade Points × Credits) / Σ(Credits).

```
 [ Full Batch Academic Report ]
-------------------------------------------------------


 Select Batch
-------------------------------------------------------
   [1] CSE2024 (Level-2 Term-2, active)
   [2] CSE2022 (Level-1 Term-1, active)
   [0] Cancel
-------------------------------------------------------

 Choice: 1

 Student : Sara Islam  (sara2024)
 Code       Course                                Credits Grade Points
-------------------------------------------------------
 CSE141     Structured Programming                3.00    A+    4.00
 EE181      Basic Electrical Engineering          3.00    A+    4.00
 MATH141    Differential and Integral Calculus    3.00    A     3.75
 PHY141     Physics I                             3.00    A+    4.00
```

Figure 21: Full Batch Academic Report

6. Activity Log

6. View Activity Log

- **Function:** Display the last 40 entries from the system audit trail.
- **Process:**
  - Reads from logs/activity.log.
  - Every login, logout, user creation, grade entry, batch operation, and registration action is recorded.
  - Columns: Timestamp, Username, Action.
  - Useful for monitoring system usage and auditing changes.

```
  [ Activity Log (last 40 entries) ]
------------------------------------------------------------

  Timestamp           User            Action
------------------------------------------------------------
  2026-04-18 12:28:25  system          FIRST_RUN_ADMIN_CREATED:admin
  2026-04-18 12:28:42  admin           LOGIN
  2026-04-18 12:29:01  admin           LOGOUT
  2026-04-18 12:29:36  admin           LOGIN
  2026-04-18 12:30:31  admin           DELETE_USER:rony2024
  2026-04-18 12:31:09  admin           ADD_USER:rhrony05:student
  2026-04-18 12:31:28  admin           LOGOUT
  2026-04-18 12:31:36  rhrony05        LOGIN
  2026-04-18 12:31:44  rhrony05        PASSWORD_CHANGED
```

Figure 22: Activity Log Viewer

III. Teacher Manual

Teacher Login

- Teacher enters their username and password.
- The password is SHA-256 hashed and verified against the stored credential in db/users.txt.
- On success, a welcome message is displayed and the Teacher Panel is loaded.
- If logging in for the first time with a temporary password assigned by the admin, the system forces a password change before granting access to the panel.

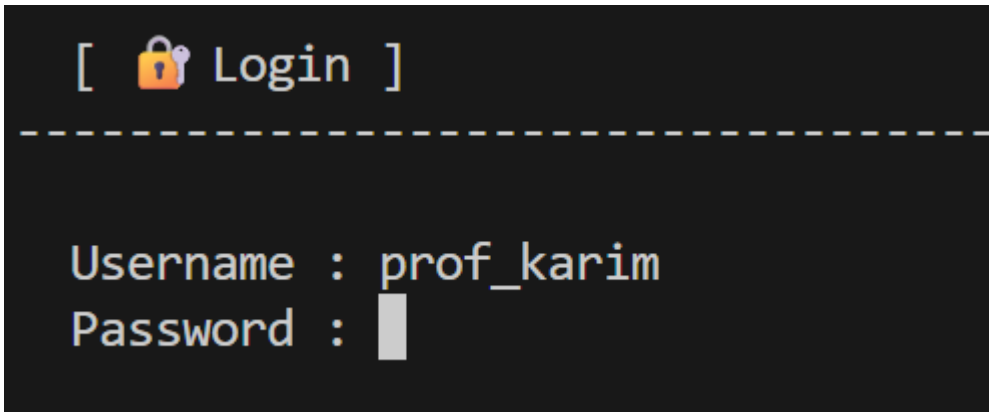

Figure 23: Teacher Login Screen

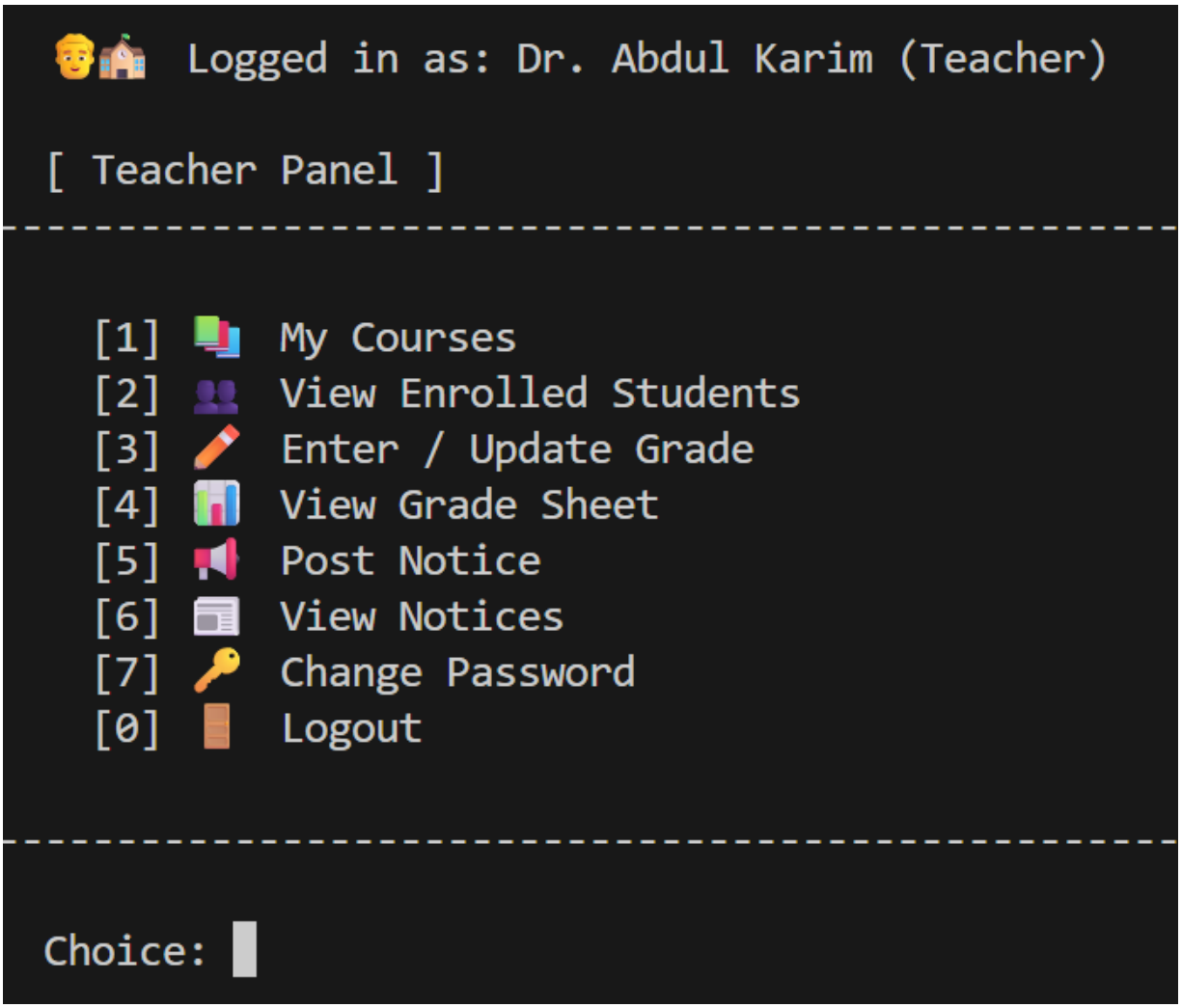

Figure 24: Teacher Panel Main Menu

1. My Courses

- **Function:** Display all courses currently assigned to the logged-in teacher.
- **Process:**
  - Reads teacher-course assignments from db/teacher_courses.txt.
  - For each assigned course, retrieves details from db/courses.txt.
  - Displays: Course Code, Course Name, Credits, Semester (Level-Term), Type.
  - If no courses are assigned, the system prompts the teacher to ask the admin for assignments.

```
[ My Courses ]
-----------------------------------------------------------

Code       Course Name                              Credits  Semester          Type
-----------------------------------------------------------
CSE141     Structured Programming                   3.00     Level-1 Term-1    theory
CSE142     Structured Programming Sessional         1.50     Level-1 Term-1    sessional
CSE143     Object Oriented Programming              3.00     Level-1 Term-2    theory
CSE144     OOP Sessional                            1.50     Level-1 Term-2    sessional
CSE241     Data Structures                          3.00     Level-2 Term-1    theory
CSE242     Data Structures Sessional                1.50     Level-2 Term-1    sessional
CSE243     Digital Electronics and Pulse Techniques 3.00     Level-2 Term-1    theory
CSE244     Digital Electronics Sessional            1.50     Level-2 Term-1    sessional
```

Figure 25: My Courses — Teacher View

2. View Enrolled Students

- **Function:** Display all students enrolled in a selected assigned course.
- **Process:**
  - Teacher selects a course from their assigned course list.
  - Reads db/enrollments.txt and filters by course code.
  - Displays: Username, Full Name, Batch, Enrollment Status.
  - Shows total enrolled student count.

```
Choice: 1

Course: [CSE141] Structured Programming

Username        Full Name                    Batch        Status
-----------------------------------------------------------
rony2024        Rony Ahmed                   CSE2024      completed
sara2024        Sara Islam                   CSE2024      completed
karim2024       Karim Hossain                CSE2024      completed

Total enrolled: 3
```

Figure 26: Enrolled Students for a Course

3. Enter / Update Grade

- **Function:** Record or update a grade for a student in an assigned course.
- **Process:**
  - Teacher selects a course from their assigned list.
  - A student picker displays all enrolled students along with their current grade (or '--' if not yet graded).

- Teacher selects a student and enters a grade from the valid set: A+, A, A-, B+, B, B-, C+, C, D, F.
- The system automatically converts the letter grade to GPA points (e.g., A+ = 4.00, B = 3.00, F = 0.00).
- If a grade already exists for the student, it is replaced with the new entry.
- The grade record is saved to courses/{CODE}/grades.txt with a timestamp.

```
  Course: [CSE241] Data Structures


  Select Student to Grade
------------------------------------------------------------
    [1] rony2024 - Rony Ahmed (CSE2024) [Grade: --]
    [2] sara2024 - Sara Islam (CSE2024) [Grade: --]
    [3] karim2024 - Karim Hossain (CSE2024) [Grade: --]
    [4] rhrony05 - Muhammad Rony (CSE2024) [Grade: --]
    [5] rafNsane - Rafsane Ron (CSE2024) [Grade: --]
    [0] Cancel
------------------------------------------------------------

  Choice:
```

Figure 27: Grade Entry Interface

4. View Grade Sheet

- **Function:** Display all graded students for a selected course with class statistics.
- **Process:**
  - Teacher selects a course.
  - Displays: Username, Batch, Full Name, Grade, GPA Points, Date of entry.
  - Calculates and displays the class average GPA points at the bottom.

```
 Course: [CSE142] Structured Programming Sessional

 Username        Batch        Full Name                  Grade   Points  Date
-------------------------------------------------------------
 rony2024        CSE2024      Rony Ahmed                 A       3.75    2026-04-18
 sara2024        CSE2024      Sara Islam                 A+      4.00    2026-04-18
 karim2024       CSE2024      Karim Hossain              B+      3.25    2026-04-18

 Graded: 3  |  Class Average Points: 3.67
```

Figure 28: Grade Sheet with Class Average

5. Post Notice

- **Function:** Publish an announcement or notice to a course's notice board.
- **Process:**
  - Teacher selects a course.
  - Teacher enters a notice title and message body.
  - The notice is saved to courses/{CODE}/notices.txt with a unique ID, author username, and timestamp.
  - Students enrolled in the course can read the notice from their panel.

```
Choice: 2
Course: [CSE142] Structured Programming Sessional

Notice Title  : Test Post Notice
Message       : What am i doing
✅ Notice posted to CSE142.
```

Figure 29: Post Notice Interface

6. View Notices

- **Function:** Read all notices posted to a selected course.
- **Process:**
  - Teacher selects a course.
  - All notices are displayed in order, each showing: Notice number, Title, Posted by, Date, and Message.

```
Choice: 2

Notices for [CSE142] Structured Programming Sessional
------------------------------------------------------------

[1] Test Post Notice
    Posted by: prof_karim on 2026-04-19
    What am i doing
```

Figure 30: View Course Notices — Teacher

7. Change Password

- **Function:** Update the teacher's own account password.
- **Process:**

- Teacher enters their current password for verification.
- Teacher enters and confirms a new password (minimum 6 characters).
- The new SHA-256 hash is saved to db/users.txt.

IV. Student Manual

Student Self-Registration

Students do not receive direct login credentials from the admin. Instead, they self-register from the system's main landing page and await admin approval.

- Student selects 'Register (Student)' from the main menu.
- Student enters a desired username (alphanumeric and underscores only, must be unique).
- Student enters their full name.
- Student selects their target batch from the batch picker.
- Student sets a password (minimum 6 characters, must be confirmed).
- Payment section: Student enters a 10-character transaction ID comprising a mix of uppercase letters and numbers (registration fee: 500 BDT).
- The registration is submitted with status 'pending' and the admin is notified via the Pending Registrations menu.
- Once approved by the admin, the student can log in immediately using the credentials they set during registration.

```
Desired Username : Shafayet47
Full Name        : Shafayet SM

ℹ️  Select your target batch:

Select Batch
------------------------------------------------------------
  [1] CSE2024 (Level-2 Term-2, active)
  [2] CSE2022 (Level-1 Term-1, active)
  [0] Cancel
------------------------------------------------------------

Choice: 2

Password       :
Confirm        :


[ 💳 Payment System ]
------------------------------------------------------------
Registration fee is 500 BDT.
Transaction ID must be exactly 10 characters,
comprising a mix of uppercase letters and numbers.

Enter Transaction ID: ASDFG12345

✅ Registration submitted successfully!
ℹ️  Your transaction ID is ASDFG12345.
ℹ️  Hold onto this ID. An admin will review your registration.
```

Figure 31: Student Self-Registration Form

Student Login

- Student enters their username and password.
- Credentials are verified against db/users.txt using SHA-256 hash comparison.
- If still pending or rejected, an appropriate message is displayed without granting access.
- On successful login, the Student Panel is loaded.

```
🎓 Logged in as: Muhammad Rony (Student)

[ Student Panel ]
-----------------------------------------------

  [1] 🏠 Dashboard
  [2] 📚 My Current Courses
  [3] 📊 My Grades (Current Semester)
  [4] 🎓 Academic Record and CGPA
  [5] 📢 Course Notices
  [6] 🔑 Change Password
  [0] 🚪 Logout

-----------------------------------------------

Choice:
```

Figure 32: Student Panel Main Menu

1. Dashboard

- **Function:** Display a summary of the student's academic standing and current status.
- **Process:**
  - Shows: Full name, username, batch, current semester (Level-Term), and batch status.
  - Calculates and displays the student's total completed credits.
  - Shows the student's current CGPA, calculated from all courses with status 'completed'.
  - If no semesters have been completed yet, displays 'No completed semesters yet'.

```
[ My Dashboard ]
-----------------------------------------------

Name     : Muhammad Rony
Username : rhrony05
Batch    : CSE2024
Semester : Level-2 Term-2
Status   : active

-----------------------------------------------

No completed semesters yet.
```

Figure 33: Student Dashboard

2. My Current Courses

- **Function:** Display all courses the student is actively enrolled in for the current semester.
- **Process:**
    - Filters db/enrollments.txt for the student's records with status 'active'.
    - Displays: Course Code, Course Name, Credits, Type (Theory/Sessional).
    - If the student has not been enrolled yet, prompts them to contact the admin.

```
[ My Current Courses — Level-2 Term-2 ]
------------------------------------------------------------

Code       Course Name                              Credits  Type
------------------------------------------------------------
CSE247     Algorithm Design and Analysis            3.00     theory
CSE249     Computer Organization and Architecture   3.00     theory
MATH243    Statistics and Probability               3.00     theory
CSE251     Theory of Computation                    3.00     theory
HUM243     Technical Writing                        2.00     theory
CSE248     Algorithm Design Sessional               1.50     sessional
CSE250     Computer Architecture Sessional          1.50     sessional

Total: 7 course(s)
```

Figure 34: My Current Courses

3. My Grades (Current Semester)

- **Function:** Display grades for the student's current active semester.
- **Process:**
  - Lists all courses in the current semester with their published grades and GPA points.
  - Courses without a grade yet show '--'.
  - Calculates Semester GPA (SGPA) from graded courses using weighted average: SGPA = Σ(Grade Points × Credits) / Σ(Credits).
  - If no grades have been published, displays 'Grades not published yet'.

```
[ My Grades — Level-2 Term-2 (Current) ]
------------------------------------------------------------

Code       Course                                   Credits  Grade  Points
------------------------------------------------------------
CSE247     Algorithm Design and Analysis            3.00     --     --
CSE249     Computer Organization and Architecture   3.00     --     --
MATH243    Statistics and Probability               3.00     --     --
CSE251     Theory of Computation                    3.00     --     --
HUM243     Technical Writing                        2.00     --     --
CSE248     Algorithm Design Sessional               1.50     --     --
CSE250     Computer Architecture Sessional          1.50     --     --

Grades not published yet.
```

Figure 35: My Grades — Current Semester with SGPA

4. Academic Record and CGPA

- **Function:** Display the student's full academic transcript across all semesters.
- **Process:**
  - Shows all enrolled semesters, both completed and active, in chronological order.
  - For each semester: lists all courses with grade and GPA points.
  - Prints the Semester GPA at the bottom of each semester block.
  - At the end of the transcript, displays total completed credits and overall CGPA.
  - CGPA is calculated from completed enrollments only: CGPA = Σ(Grade Points × Credits) / Σ(Credits).

```
  [ Full Academic Record ]
-------------------------------------------------------------
  Student : Muhammad Rony  (rhrony05)
  Batch   : CSE2024

  ── Level-2 Term-1  [completed] ──
  Code       Course                                  Credits  Grade  Points
-------------------------------------------------------------
  CSE241     Data Structures                         3.00     --     --
  CSE243     Digital Electronics and Pulse Techniques 3.00     --     --
  MATH241    Complex Variable, Laplace and Fourier   3.00     --     --
  CSE245     Discrete Mathematics                    3.00     --     --
  HUM241     Sociology                               2.00     --     --
  CSE242     Data Structures Sessional               1.50     --     --
  CSE244     Digital Electronics Sessional           1.50     --     --
  MATH242    Mathematics Sessional                   0.75     --     --

  ── Level-2 Term-2  [active] ──
  Code       Course                                  Credits  Grade  Points
-------------------------------------------------------------
  CSE247     Algorithm Design and Analysis           3.00     --     --
  CSE249     Computer Organization and Architecture 3.00     --     --
```

Figure 36: Full Academic Record and CGPA Transcript

5. Course Notices

- **Function:** Read announcements posted by teachers for enrolled courses.
- **Process:**
  - Student selects a course from all their enrollments (current and past).
  - All notices for that course are displayed in order: Notice number, Title, Date, Posted By, and Message.
  - Students can only read notices — they cannot post or delete them.

```
  Notices for [CSE241] Data Structures
------------------------------------------------------------

  [1] Welcome to Data Structures
      Date   : 2026-04-18
      From   : prof_karim
      Welcome to CSE241. Course materials will be shared on the portal.

  [2] Assignment 1 Released
      Date   : 2026-04-18
      From   : prof_karim
      Assignment 1 on Linked Lists is due by end of this week.

  [3] Welcome to Data Structures
      Date   : 2026-04-18
      From   : prof_karim
      Welcome to CSE241. Course materials will be shared on the portal.
```

Figure 37: Course Notices — Student View

6. Change Password

- **Function:** Update the student's own account password.
- **Process:**
  - Student enters their current password for verification.
  - Student enters and confirms a new password (minimum 6 characters).
  - The new SHA-256 hash is saved to db/users.txt.

Conclusion

By completing the CUET CSE Student Management System project, we gained practical experience in designing and implementing a multi-role, file-based information management system using only Bash shell scripting — without relying on any external database or web framework. This experience has deepened our understanding of how operating system-level tools can be used to build functional, real-world applications.

The functionalities implemented — including student self-registration with payment verification, role-based access control for Admin, Teacher, and Student roles, multi-semester enrollment tracking, grade management with automatic SGPA and CGPA calculation, a course notice board, and a full activity audit log — collectively simulate the operational needs of a university academic records system.

This project also strengthened our skills in modular Bash scripting, SHA-256-based password hashing, flat-file database design using pipe-delimited records, in-place file updates using awk and temporary files, and terminal UI design with menus and interactive pickers. It demonstrates that even a lightweight, shell-based system can handle complex business logic and provide a reliable user experience for managing academic records at a departmental level.